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GAME TOKEN MANAGEMENT SYSTEM

Abstract

A controller of the table game management system is configured to be able to recognize that a bet chip bet by a winning player and a payout chip paid to the winning player are transferred outside a chip placement area, and associate, in a database, ID information of the payout chip with ID information of the bet chip transferred outside the chip placement area at the same timing as the payout chip, and includes a function to associate, in the database, ID information held by RFID of the payout chip with information about an owner of the bet chip indicated by RFID of the bet chip.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/408,926 filed Jan. 10, 2024, which is a continuation of U.S. patent application Ser. No. 17/679,523 filed Feb. 24, 2022 (now abandoned), which is a continuation of U.S. patent application Ser. No. 16/877,892 filed May 19, 2020 (now U.S. Pat. No. 11,288,923), which claims priority to JP Pat. App. No. 2019-106546 filed May 20, 2019. The entire contents of each application are hereby incorporated by reference.

DESCRIPTION

Field

[0002] The present disclosure relates to a game token management system that prevents a wrong game token from being used at a gaming hall in a casino or another gaming facility that uses game tokens.

Background Art

[0003] There are baccarat and blackjack among many table games played in a gaming hall such as a casino. In a casino, game tokens are used for such games.

[0004] Use of a wrong game token including a counterfeit game token and a broken game token is not allowed in a casino. In order to prevent use of a counterfeit game token in a casino, casinos have technology to determine authenticity of game tokens with a radio frequency identification (RFID) tag attached for preventing use of a counterfeit game token. A game token with RFID is known and disclosed in International Application Publication No. WO 2008/120749.

[0005] With a game token based on conventional technology, it has been possible to determine authenticity of the game token itself by determining information written in a memory area by reading RFID. However, information obtained during use of the game token at a casino site, in particular, information about an owner of the game token has not been able to be recorded and determined. Therefore, it has not been possible to determine, by reading RFID, presence of a genuine game token having abnormal history in terms of usage, such as history of being used or exchanged by an owner different from an original owner, failing to prevent use of such a game token having strange use history in a casino.

[0006] Therefore, it is desired to provide a system that makes it possible to determine and manage owner information of each game token.

SUMMARY

[0007] In order to solve the above-described conventional problem, the game token management system in one aspect includes a game token to which individually identifiable RFID is attached, a game table for playing a game by using the game token, including a chip placement area on which the game token is placed as a bet chip that is bet by a player participating in a game at the game table and as a payout chip that is paid out from a chip tray of the game table by a dealer to a winning player, a chip reader that reads the RFID of the game token placed on the chip placement area, and a controller that determines a reading result by the chip reader, in which the controller is able to acquire owner information of the game token on the basis of identification (ID) information of the RFID of the game token, the controller is configured to be able to recognize that the bet chip of a winning player and the payout chip paid out to the winning player are transferred to outside the chip placement area, and associate, in a database, ID information of the payout chip with ID information of the bet chip transferred to outside of the chip placement area at the same timing as

the payout chip, and the controller includes a function to associate, in the database, owner information of the bet chip indicated by the RFID of the bet chip to ID information held by the RFID of the payout chip.

[0008] Further, the chip placement area may be divided for each player position and may include a payment area for placing the payout chip for each player position. The chip reader may be configured to be able to read the payment area separately for each player position.

[0009] The foregoing and other objects, features, aspects and advantages of the exemplary embodiments will become more apparent from the following detailed description of the exemplary embodiments when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a diagram of an overall overview of a table game management system according to an embodiment.

[0011] FIG. 2 is a diagram of an overview of a database according to the embodiment.

[0012] FIG. 3 is an explanatory diagram of an RFID tag included in a game token according to a second embodiment.

[0013] FIG. 4 is a conceptual diagram of a method for recording data of a game token according to the second embodiment.

[0014] FIG. 5 is a diagram of an overall overview of a game token management system according to the second embodiment.

[0015] FIG. 6 is a diagram of a reader and writer of the game token according to the second embodiment.

[0016] FIG. 7A is an explanatory diagram of configuration of data of the game token according to the second embodiment.

[0017] FIG. 7B is an explanatory diagram of configuration of data of a game token according to another second embodiment.

[0018] FIG. 8 is a diagram of a database according to the other second embodiment.

[0019] FIG. 9 is a diagram of an overall overview of a table game management system at a gaming hall having a plurality of game tables according to the embodiment.

[0020] FIG. 10 is a diagram of a result of reading a chip by a camera and RFID according to the embodiment.

[0021] FIG. 11 is a diagram of a game table according to the embodiment.

[0022] FIG. 12 is a diagram of a state of game tokens bet at the game table according to the embodiment.

[0023] FIG. 13 is a diagram of a state of game tokens bet at the game table according to the embodiment.

[0024] FIG. 14A is a diagram of a detail of the chip tray according to a first embodiment.

[0025] FIG. 14B is a diagram of another example of a chip tray according to the first embodiment.

[0026] FIG. 15 is a diagram of the game table according to the embodiment.

[0027] FIG. 16 is a diagram of a result of reading a chip by the camera and RFID according to the embodiment.

[0028] FIG. 17 is a diagram of a game table according to another embodiment.

[0029] FIG. 18 is a diagram of a game table according to the other embodiment.

[0030] FIG. 19 is a diagram of a game table according to the other embodiment.

[0031] FIG. 20 is a perspective view of a game token according to the embodiment.

[0032] FIG. 21 is a cross-sectional view of the game token according to the embodiment.

[0033] FIG. 22 is a cross-sectional view of the game token according to the embodiment.

[0034] An overall overview of a table game management system according to a first embodiment will be described with reference to FIG. 1. FIG. 1 is a diagram of an overall overview of the table game management system. A game table **1004** for a baccarat game includes betting areas including player **1441**, banker **1442**, tie **1443**, player pair **1444**, and banker pair **1445** for each of player positions **1131** to **1135**. Each of the player positions is provided with a chip placement area **1025**. The table game management system further includes a chip reader **1002** that reads RFID of game tokens **1001** placed on the chip placement areas.

[0035] Each of the chip placement areas **1025** includes a payment area **1026** on which a dealer places a game token **1001** removed from a chip tray **1009** of the game table **1004** when the dealer pays the game token **1001** out of the chip tray **1009** to a winning player.

[0036] Further, the table game management system includes a controller **1003** that determines a result of reading RFID of the game token **1001** read by the chip reader **1002**.

[0037] First, a player who participates in a game bets a game token **1001** on a betting area in the chip placement area **1025**. By the chip reader **1002** reading RFID of the bet game token **1001**, the controller **1003** determines owner information of the game token **1001**. The owner information is associated with ID information of the RFID in a database.

[0038] In a case where a casino wins based on a result of the game, the dealer collects a game token **1001** bet by a losing player and places the game token **1001** in the chip tray **1009**. The chip reader **1002** determines RFIDs of game tokens **1001** in the chip tray **1009**, and the controller **1003** records in the database that the game token **1001** collected by the casino is owned by the casino.

[0039] If a player wins based on the result of the game, the dealer pays a game token **1001** in an amount corresponding to an amount of a bet chip out of a chip tray to the winning player according to a rule of the game.

[0040] At a time of payout, a payout chip may be placed on the above-described payment area **1026** or on a chip placement area **1025** other than the payment area.

[0041] Here, a method for recording owner information of the payout chip will be described with reference to FIG. 2. FIG. 2 illustrates a method for recording owner information of the game token **1001** when the chip is paid to the winning player.

[0042] In a case where a plurality of players place bet chips, and there are two or more stacks of the bet chips in the same player position, the chip reader **1002** collectively reads the game tokens **1001** placed on the chip placement area **1025**. Thus, during a game, the chip reader **1002** acquires an ID **1010** of the bet game tokens without determining which game token **1001** belongs to which player. When payment is to be made to a winning player after determination of a win/loss result of the game, the dealer pays out to each winning player (for each stack) from the chip tray **1009**. A game token **1001** removed from the chip tray **1009** is placed on a chip placement area **1025**. The chip reader **1002** reads an ID **1011** of the payout chip placed on the chip placement area **1025**.

[0043] The winning player receives the paid out game token **1001** together with the originally bet game token **1001**. At this time, because the bet chip and the payout chip transfer outside the chip placement area **1025**, an ID of the bet chip and an ID of the payout chip disappear from the reading result by the chip reader **1002**. With this arrangement, the controller **1003** determines to which player the payout chip has been paid by associating with the owner information of the bet chip.

[0044] The controller **1003** stores the determined owner information of the payout chip in the database by associating the owner information with the ID of the game token.

[0045] The above-described arrangement enables traceability of the owner of the game token **1001** and, in a case of wrong transfer of the game token **1001**, detection of the owner information stored in the database being different from information of an actual owner.

[0046] Note that the actual owner can be identified by using face authentication technology or an ID of a membership card.

[0047] In the above description, an ID of each game token is managed in a database. However, the

game token **1001** itself may store owner information.

[0048] In addition to the above-described case where a payment area **1026** is provided to each player position, a payment area may be provided separately from the player positions, and the chip reader **1002** may perform reading independently.

[0049] Use or non-use of the payment area **1026** may be selected depending on reading performance of the chip reader **1002**.

[0050] The table game management system may be able to use information mutually with an RFID reader that reads chips in the chip tray **1009** by RFID, a camera that reads a bet chip, a win/loss result determination device such as a shoe that determines a win/loss result of a game, or other device, which is connected to the table game management system.

[0051] First, a game token used in a game token management system according to an embodiment will be described. FIG. **3** illustrates a game token **1** used in the game token management system. In FIG. **3**, an RFID tag **2** capable of storing various pieces of information is embedded in the game token **1**. The RFID tag **2** has a data non-rewritable area **21** and a data rewritable area **22**.

[0052] The data non-rewritable area **21** stores, as fixed information **3**, information that is not to be changed, that does not change, or that must not be changed while the game token **1** is used. Specifically, the data non-rewritable area **21** stores, as the fixed information **3**, production information, product information, casino information, amount information, a serial number, or the like of the game token **1**. The production information includes date and time when the game token was manufactured, manufacturing machine used for manufacturing the game token, or other information. The product information includes, for example, information indicating that the game token is a chip for a VIP area of a casino and information indicating a type of the chip (for example, rolling chip or cash chip).

[0053] The data non-rewritable area **21** may be a functionally non-writable area due to a specification of the RFID tag, or may be a writable area locked to disable writing of data after necessary information is written.

[0054] The data rewritable area **22** stores, as variable information **4**, information that changes while the game token **1** is used. For example, as illustrated in FIG. **3**, the data rewritable area **22** stores, as the variable information **4**, location information **220** including date/time information **221**, place/event information **222**, owner information **223**, or other information.

[0055] The fixed information **3** and the variable information **4** may be encrypted in order to prevent wrong reading or writing of information by another person. Further, the fixed information **3** and the variable information **4** may be stored as metadata.

[0056] FIG. **4** illustrates an example of a method for recording data of the variable information **4** of the game token **1**. As illustrated in FIG. **4**, the location information **220** including date/time information **221**, place/event information **222**, and owner information **223** is set as a block. Then, blocks of the location information **220** are connected in a chain so that history of the location information **220** of the game token **1** can be determined.

[0057] FIG. **4** illustrates an example of the variable information **4** stored in the game token **1**, and transference of the game token **1** as described below can be determined. At 16:02 on Jan. 28, 2019, Mr. A exchanges cash and a game token **1** at a cage **5**. At 16:15 on the same day, Mr. A makes a bet at table No. 325 by using the game token **1**. At 16:43 on the same day, Mr. B receives the game token **1** as a redemption for having won a bet at table No. 325. At 17:01 on the same day, Mr. B exits a casino, carrying the game token **1**.

[0058] The variable information **4** may be configured to store only latest location information as illustrated in FIG. **4** or store all written location information. Further, some of a plurality of pieces of location information may be selected and stored. In a case where some pieces of the location information **220** are selected and stored among all of the location information, a plurality of pieces of the location information including at least the latest location information **220** may be stored. With the above configuration, history of use or transference of a certain game token **1** is known

from variable information **4** of the game token **1**. That is, traceability information of the game token **1** is written in the game token **1** itself to be determined.

[0059] Next, a management system for the game token **1** of the present embodiment will be described. FIG. **5** illustrates an overall overview of the management system.

[0060] First, at a factory **6**, a writer **14** writes, in the data rewritable area **22** of the RFID tag **2** of the game token **1**, the variable information **4** including information of manufacturing completion or factory shipment as location information **220**.

[0061] The game token **1** shipped from the factory **6** is subjected to a receiving process in a backyard **7** of the casino. In the backyard **7**, a reader **13** reads the variable information **4** stored in the data rewritable area **22** of the game token **1**. Then, on the basis of the reading result, a management controller **15** determines whether or not the location information **220** indicating that the information has been written at the factory **6**, which is expected to be written, is actually written as latest variable information **4**. If there is no record of writing at the factory **6**, an error signal is generated indicating that the game token is possibly being wrong. This allows the casino to refuse to receive the game token or to request a factory to conduct investigation. If there is no problem in record of writing, in the data rewritable area **22** of the RFID tag **2**, the writer **14** writes location information **220** indicating a backyard. At a time of writing, the location information indicating the backyard may be written either in addition to the location information indicating the factory or after deleting the location information indicating the factory. Further, reading and writing of the game token in the backyard **7** described above can be performed together with usual validation or activation of the game token.

[0062] The game token **1** of which receiving inspection has been completed in the backyard **7** is carried to a warehouse **8** or cage **9** of the casino. In the warehouse **8** or the cage **9**, the reader **13** reads the variable information **4** stored in the data rewritable area **22** of the game token **1**. Then, on the basis of a reading result, a management controller **15** determines whether or not the location information **220** indicating that the information has been written at the backyard **7**, which is expected to be written, is actually written as latest variable information **4**. Further, it may be also inspected whether information indicating the factory **6** is written in history of the variable information **4**. The management controller **15** determines whether or not any abnormality is present in the history of the location information, similarly to the above-described inspection in the backyard **7**. If there is no problem in a reading result, the writer **14** writes, in the data rewritable area **22**, location information **220** indicating the warehouse **8** or the cage **9**.

[0063] A player purchases the game token **1** at the cage **9**. When the player purchases the game token **1**, a writer **14** at the cage **9** writes, as the variable information **4**, time at which the game token is purchased and information indicating change of the owner of the game token from the casino to a customer. The player may be identified and recorded by face authentication technology or an ID card such as a membership card of the casino or an official ID card.

[0064] The player makes a bet by placing the purchased game token **1** on a betting area **11** at a game table. From the RFID tag **2** of the bet game token **1**, a reader **13** reads the location information **220** including date/time information **221**, place/event information **222**, and owner information **223**. The management controller **15** determines whether or not any abnormality is present. It can be determined that abnormality is present, for example, in a case where a certain period has passed since last written date and time of information of use at a game table or information of cashing in at a cage, or in a case where the owner information **223** stored in the game token **1** and the owner information **223** identified by face authentication or an ID card are different from each other.

[0065] If there is no problem in a reading result, the writer **14** adds the location information **220** to the data rewritable area **22** of the bet game token **1**. The location information **220** includes date/time information **221**, place/event information **222**, and owner information **223**. At the table, for example, the writer **14** writes variable information **4** including information of a player position

number where the game token **1** is placed and information of the player. The player information may be identified by an ID card or face authentication. Usually, a plurality of game tokens **1** are stacked and placed on the betting area **11**, and therefore writing is performed collectively for the stacked game tokens **1**.

[0066] When the player takes the game token **1** out of a casino floor, exit processing is performed at a gate **10**. At a time of exit of the player, the reader **13** reads the location information **220**, and the management controller **15** determines whether or not any abnormality is present with the variable information **4**. Further, at the gate **10**, the writer **14** writes the variable information **4** including place/event information **222** indicating the exit and owner information **223**.

[0067] In a case where the player brings in the game token **1**, which has been taken out before, and enters the casino floor, the reader **13** reads the location information, and a management controller determines whether or not any abnormality is present with the variable information **4**. It is possible to determine that there has been transfer of the game token outside the casino, for example in a case where the owner of the game token **1** once taken out of the casino is different between at the time of exit the casino and at the time of entry to the casino.

[0068] The above-described management controller **15** may output an alarm to the cage **9** or to the gate **10** in a case where a reading result by the reader **13** is determined to be abnormal. On the basis of the alarm output by the management controller **15**, cashing in may be refused when the owner of the game token **1** attempts to cash in at the cage **9**, or a check may be conducted individually when the owner of the game token **1** exits the gate **10**. Further, in a case where a determination result is abnormal, the management controller **15** may issue an alarm to a dealer of each game table to interrupt a game or may issue an alarm to a pit, and a pit manager may determine to interrupt or continue a game, or determine to exchange the game token **1**. Alternatively, the management controller **15** may be connected to an overall management controller **18** of the casino and may issue an alarm to the overall management controller **18**.

[0069] Similarly, the management controller **15** may determine whether or not any abnormality is present in the variable information **4** read by the reader **13** when the game token **1** is exchanged for cash at the cage. Specifically, the management controller **15** may determine that abnormality is present in a case where a certain period has passed since latest written date and time of information of use at a game table or information of cashing in at the cage, or in a case where the owner information **223** stored in the game token **1** and information of a person who is going to exchange the game token **1** for cash are different from each other. In a case where abnormality has been determined to be present with the game token **1**, the management controller **15** may output an alarm to refuse to exchange the game token **1** for cash.

[0070] Further, the above description is given of a case where the reader **13** performs reading, a management controller performs determination, and then the writer **14** performs writing. However, the reading and the writing may be performed simultaneously.

[0071] In the following, details of application of an embodiment will be described.

[0072] At the factory **6**, the RFID tag **2** is read and/or written at a time of production completion or product shipment. The production information and the product information are written as the fixed information **3** and locked as appropriate so as not to be rewritten, and the variable information **4** is written.

[0073] At the backyard **6**, the game token **1** shipped from the factory is received, and the RFID tag **2** is read and/or written when the game token **1** is activated as a usable game token **1**.

[0074] At the warehouse **8**, the RFID tag **2** is read and/or written when the game token **1** is transferred from the warehouse **8** to the cage **9**, or from the cage **9** to the warehouse **8**.

Alternatively, the RFID tag **2** of the game token **1** kept in the warehouse **8** may be read and/or written at regular time intervals or at a predetermined timing.

[0075] At the cage **9**, the RFID tag **2** is read and/or written when the game token **1** is transferred from or to the warehouse **8** or when cash owned by a customer and the game token **1** are

exchanged. In a case where cash and the game token **1** are exchanged at the cage **9**, as the variable information **4**, place/event information **222** indicating the exchange and owner information **223** indicating a player who exchanged the cash and the game token **1** are written. The owner information **223** can be acquired from a casino ID card of the player, a face authentication system, a credit card, or the like.

[0076] At the gate **10**, the RFID tag **2** is read and/or written on entry or exit of the player. To all game tokens **1** that a player takes out of the casino when exiting the casino, as the variable information **4**, place/event information **222** indicating the taking out and owner information **223** indicating a player who takes out the game tokens **1** are written and registered. Similarly, at a time of entry to the casino, as the variable information **4**, place/event information **222** indicating the bringing in and owner information **223** indicating a player who brings in the game token **1** are written and registered to the game token **1** brought in the casino.

[0077] At the betting area **11**, the RFID tag **2** of the game token **1** that a player placed on the betting area **11** to participate in a betting is read and/or written. The RFID tag **2** of a game token **1** that the dealer has placed on a betting area **11** as redemption to the player is read and/or written.

[0078] In a chip tray **12**, the RFID tag **2** of each of the game tokens **1** collected in the chip tray and the RFID tag **2** of each of the game tokens **1** kept in the chip tray are read and/or written.

[0079] Further, the game table may include a payout area including area **2007** and the payment area **1026**, and the RFID tag **2** of a game token **1** that the dealer has placed on the betting area **11** as redemption to a player may be read and/or written.

[0080] The management controller **15** has a function to determine whether or not any abnormality is present in location information **220** obtained from a reading result by a reader **13** at each place. Determination of presence of abnormality is made in a case where the game token **1** has not been used for a predetermined period since last writing, or where location information, which is supported to be written, is not written.

[0081] A situation where the game token **1** has not been used for a predetermined period since last writing is a situation where: 1) latest location information indicates the cage **9**, and a predetermined period has passed before next information is written; 2) latest location information indicates entry at the gate **10**, and a predetermined period has passed before next information is written; or 3) latest location information indicates exit at the gate **10**, and a predetermined period has passed before next information is written.

[0082] A situation where location information, which is supported to be written, is not written is a situation where: 4) at a time of entry to a gaming hall, information indicating exit at the gate **10** is not written as latest location information in data rewritable area **22** of the game token **1**; 5) at a time of use at the cage **9**, information indicating the backyard **7** is not written; or 6) at a time of exit from the gaming hall, information indicating the cage **9** or the backyard **7** is not written.

[0083] On the basis of the owner information **223** obtained from the reader **13**, the management controller **15** can determine presence of abnormality when: 1) a person different from a latest owner whose information is stored in the data rewritable area **22** exchanges the game token **1** for cash or another game token in the gaming hall, or exits the gaming hall; or 2) a person different from an owner at a time of exit whose information is stored in the data rewritable area **22** brings the game token in the gaming hall.

[0084] On the basis of information obtained from the reader **13** and indicating a place of the betting area **11**, or information indicating location of the chip tray **12** or a payout area, the management controller **15** determines presence of abnormality when: 1) a person different from a latest owner whose information is stored in the data rewritable area **22** uses the game token **1** at a game table; 2) a person different from the person who purchased, at the cage **9**, a game token **1** without a record of use at the game table attempts to exchange the game token **1** at the cage **9**; 3) a game token **1** not including information indicating the cage **9** or the backyard **7** is used in the gaming hall; or 4) latest location information is information indicating the betting area **11** or the payout area, and after a

predetermined period has passed, the game token **1** is exchanged for cash at the cage **9** or is used at the game table.

[0085] FIG. **8** illustrates a database according to another embodiment. The management system further includes a database **17** that records similar fixed information **3** and variable information **4** outside the game token **1**. The management controller **15** is able to record information in the database **17** on the basis of a reading result by the reader **13**, check the fixed information **3** and the variable information **4** stored in the game token **1** with information in the database, and determine presence of abnormality.

[0086] In the RFID tag **2** according to the embodiment, a functionally non-writable area in the data non-rewritable area **21** may be a tag identifier (TID) area. An area in the data non-rewritable area **21**, which is locked to disable further writing of data after necessary information is written, may be an electronic product code (EPC) area or a user area. The data rewritable area **22** may be an EPC area or a user area.

[0087] The reader **13** and the writer **14** may be changed in form depending on a place. For example, the reader **13** and the writer **14** may be on a board at the factory **6** as illustrated in FIG. **6**, and may be box-shaped at the gate **10**. The betting area **11** or the chip tray **12** may include functions of the reader **13** and the writer **14**. Further, the game token **1** either may be read and written in a stacked state without any support or may be read and written while being contained, for example, in a chip case. Further, the reader **13** and the writer **14** may be unified.

[0088] The above embodiment describes a case where a game token has one RFID tag, and the single RFID tag has an area in which writing to data is impossible and a data rewritable area. Meanwhile, there may be a case where two RFID tags are embedded in the game token. In this case, one RFID tag stores fixed information and then is locked to disable data writing, and another RFID tag is left data-rewritable to record variable information.

[0089] However, in this case, when a plurality of game tokens are read collectively, an information group of fixed information **3** and an information group of variable information **4** are read separately as illustrated in FIG. **7A**. Therefore, a problem arises in that a combination of fixed information and variable information of a specific game token cannot be determined, and a game token having abnormal variable information cannot be identified.

[0090] On the other hand, in the case of one RFID tag, as illustrated in FIG. **7B**, mutually corresponding fixed information and variable information that are read by a reader can be determined even when a plurality of game tokens are read collectively. Therefore, a game token having a single RFID tag is conceived to be preferable.

[0091] In general, an RFID tag with a larger diameter has higher reading accuracy. Therefore, in a case where an RFID tag is embedded in the game token, a diameter of the RFID tag is preferably at least equal to or larger than a radius of the RFID tag. Thus, it is not preferable that two RFID tags are embedded in the game token, in which case, the diameter of each of the RFID tags becomes smaller. Further, the number of RFID tags to be read increases with a game token including two RFID tags, resulting in lower reading speed at a time of reading a plurality of game tokens.

[0092] In view of the above, a game token having a single RFID tag is considered to be more effective.

[0093] Described below is an overall overview of a table game management system at a gaming hall having a plurality of game tables according to an embodiment. FIG. **9** is a diagram illustrating an overall overview of the table game management system. The table game management system having a plurality of game tables **2004** at a gaming hall includes a game recorder **2011** that records an image of a game at a game table **2004** including an image of the game token **1**, via a plurality of cameras **2002**, an image analyzer **2012** that performs image analysis on the recorded images, a chip reader **25** that reads RFID added to the game token **1**, and a game result determination device **18** that determines and displays a result of each game played at the game table **2004**.

[0094] The game result determination device **18** is, for example, a card distributor **2003**, which is a

so-called electronic shoe that have been used by persons skilled in the art. A rule of a game is previously programmed in the card distributor **2003**, which is configured to read information of a distributed card C and determine win or loss of the game. For example, in a baccarat game, a win of a banker, a win of a player, and a tie (draw) are basically determined by each rank of two to three cards, and a determination result (win/loss result) is displayed by a result display lamp **2013**.

[0095] In FIG. **9**, a game table **2004** is a table for baccarat and provided with five sitting numbers **13**, allowing five game participants to participate in a game. The five sitting numbers are “sitting number **1**” **131**, “sitting number **2**” **132**, “sitting number **3**” **133**, “sitting number **5**” **134**, and “sitting number **6**” **135**. A betting area **44** is provided to each sitting number **13**. The betting area **44** of a sitting number **13** includes each area for player (PLAYER) **441**, banker (BANKER) **442**, tie (TIE) **443**, player pair (P) **444**, and banker pair (B) **445**. Each game participant makes a bet by placing a game token **1** in an amount to be bet at a position corresponding to a bet target. In FIG. **9**, for example, at a sitting number **5**, two game tokens **1** are placed on tie **443** and two game tokens **1** are placed on player **441** for betting. The game table **2004** further includes a player card area **45** for placing a drawn player card and a banker card area **46** for placing a drawn banker card.

[0096] Via a camera **2002** and the image analyzer **2012**, the controller **2014** is able to determine on which position (player, banker, tie, or pair) in a betting area **2008** each player **2006** has bet a game token **1**, and types and the number of the bet game tokens **1**. Different colors are assigned to game tokens **1** in different values. That is, it is possible to determine that, for example, at the sitting number **5**, two game tokens **1** are bet on tie **443** and two game tokens **1** are bet on player **441**.

[0097] As illustrated in FIG. **9**, the chip reader **25** is arranged so that RFIDs of the game tokens **1** in a reading area can be collectively read by regarding an entire betting area for each sitting number as one reading area. In the RFID attached to a game token **1**, unique ID information, and an amount, type, manufacturing information, owner information, or the like, of the game token **1** are written. Alternatively, the ID information and the amount, type, manufacturing information, owner information, or the like, of the game token **1** may be associated and stored in a database. By reading the RFID attached to the game token **1**, the chip reader **25** can determine a total amount or the number of game tokens **1** for each sitting number. That is, for example, a type, an amount, and an ID of each of four game tokens bet at the sitting number **5** can be determined.

[0098] As illustrated in FIG. **10**, on the basis of types and the number of the game tokens **1** bet at each betting area of each sitting number based on an image analysis result by the image analyzer **2012**, the controller **2014** determines a total amount or the total number of the game tokens **1** for each sitting number. Then, to determine agreement, the controller **2014** compares the determined a total amount or the total number of the game tokens **1** with a total amount or the total number of the game tokens **1** for each sitting number based on information of RFIDs of the game tokens **1** read by the chip reader **25**. Furthermore, a sitting number for which an amount or the total number of the game tokens **1** determined to be disagreeing can be output, and therefore a sitting number having a problem can be identified.

[0099] In a case where the total amount or the total number of the game tokens **1** for each sitting number based on the image analysis result by the chip reader **25** is smaller than the total amount or the total number of the game tokens **1** for each sitting number based on the reading result by an image analyzer, there is a possibility that RFID attached to a game token **1** is broken, the RFID cannot be read, or RFID is not attached to the game token **1**. In FIG. **10**, the read total number of game tokens **1** bet at sitting number **1**, based on image analysis by the image analyzer **2012**, is four, whereas the total number of the game tokens **1** based on reading of RFID by the chip reader **25** is three. Therefore, it is conceivable that one of the four game tokens **1** bet at the sitting number **1** has unreadable RFID.

[0100] In a case where the total amount or the total number of the game tokens **1** for each sitting number based on the image analysis result by the chip reader **25** is larger than the total amount or the total number of the game tokens **1** for each sitting number based on the reading result by the

image analyzer, there is a possibility that the image analyzer **2012** has not been able to read the game token **1** for such reasons that the game token **1** is in a blind spot of a camera. In FIG. **10**, the read total number of game tokens **1** bet at sitting number **3**, based on image analysis by the image analyzer **2012**, is one, whereas the total number of the game tokens **1** based on reading of RFID by the chip reader **25** is two. Therefore, there is a possibility that only one of the two game tokens **1** bet at a sitting number **3** has been recognized by the image analyzer.

[0101] ID information of the RFID attached to the game token **1** is registered in a database of a casino. Information of production and shipment at a factory, and information of reception, validation, and activation at the casino are recorded. The controller **2014** can check the information of the RFID read by the chip reader **25** with information in the database and determine authenticity of the registered information of the game token **1** to be used in the casino.

[0102] With this configuration, types and the number of the game tokens **1** bet at each betting area can be determined by the camera. Further, on the basis of RFID information of game tokens **1** read by the chip reader **25**, a wrong or defective game token **1** can be identified for each sitting number.

[0103] The table game management system further includes a game participant identification unit using a face authentication system or a player ID card, and the controller **2014** can associate game participant information identified by the game participant identification unit with a sitting number. For each game participant, the controller **2014** can store the bet amount, redeeming or collecting amount, and win/loss information, which are stored for each sitting number. Further, the controller **2014** can collectively store information of a plurality of games that game participants play at a plurality of tables.

[0104] By using a chip tray determination device **12**, the controller **2014** can determine a total amount of the game tokens **1** in a chip tray **17** of a dealer **2005** at the game table **2004**. After a game end and settlement, the controller **2014** can calculate to figure out whether a total amount of the game tokens **1** in the chip tray **17** has increased or decreased by comparing an amount of collected lost game tokens **1** that each player **2006** has bet with an amount of payment **9W** to a winning player **2006W** for won chips, according to a win/loss result of the game. A chip tray determination device may capture an image of the game tokens **1** contained in the chip tray **17** by using the camera **2002** and, on the basis of analysis by an image analyzer, determine the total amount of the game tokens **1** in the chip tray **17**. Further, the chip tray determination device may detect the total amount of the game tokens **1** contained in the chip tray **17** by providing an RFID reader to the chip tray **17**.

[0105] In this example, a wrong or an error is detected on the basis of a win/loss result of the game, information about how many and what type of game token **1** has been bet on which position (player, banker, tie, or pair) in the betting area **2008**, and increase or decrease in amount of the game tokens **1** in the chip tray **17** after collection of lost chips and redemption for a won game token **1**. Therefore, a wrong or an error can be detected without determining transference of the game tokens **1** after a game end, that is, whether a bet game token **1** has transferred to a player side or a dealer side.

[0106] Here, in a case of baccarat for example, a win/loss result of a game can be determined according to a rule of baccarat by the card distributor **2003** reading a rank of the card **C** drawn out in the game. Further, the win/loss result can be determined by capturing an image of a top of the game table **2004** by a camera **2002**, analyzing the image by the image analyzer **2012**, and then checking an analysis result with a rule of the game by the controller **2014**. In this case, a game result determination device includes the camera **2002**, the image analyzer **2012**, and the controller **2014**. Information of how many and what type of game tokens **1** a player at each sitting number **7** has bet on which position (player, banker, tie, player pair, banker pair) in a betting area **2008** is obtained by the camera **2002** capturing an image of the game tokens **1** placed on the betting area **2008** and the image analyzer **2012** analyzing the image for each play position **7**.

[0107] Further, increase in amount of the game tokens **1** in the chip tray **17** from before collection

to after collection of a lost game token **1** and decrease in amount of the game tokens **1** in the chip tray **17** from before redemption to after redemption for a won game token **1** can be calculated by comparison of a total amount of the game tokens **1** in the chip tray **17** before the collection of the lost game token **1** and redemption for the won game token **1** with a total amount of the game tokens **1** in the chip tray **17** after the collection of the lost game token **1** and redemption for the won game token **1**.

[0108] For example, assuming that a total amount of the game tokens **1** in the chip tray **17** before start of the game is Bb , and a total amount of the game tokens **1** in the chip tray **17** after the end of the game and collection of lost chips and redemption for a won chip are completed is Ba . Further, assuming that, in this game, a total amount of the game tokens **1** bet on player areas of all play positions **7** is bp , a total amount of the game tokens **1** bet on banker areas of all the play positions **7** is bb , and a total amount of the game tokens **1** bet on tie areas of all the play positions **7** is bt . For example, in a case where a banker is a winner as a win/loss result of the game, $Ba - Bb = bp - bb + bt$ should be established. Alternatively, Ba , which is the total amount of the game tokens **1** in the chip tray **17** after the end of the game, should be $(Bb + bp - bb + bt)$. If this is not the case, it can be determined that there has been a wrong or an error in collection or redemption of the chips.

[0109] FIG. **14A** is a diagram illustrating details of a chip tray of the present embodiment, and FIG. **14B** is a diagram illustrating another example of the chip tray. The chip tray **17** includes a collection chip tray **171** for temporarily keeping a collected game token **1L** bet by a losing player **2006L**, and a redemption chip tray **172** for keeping a game token **1W** to be redeemed. The image analyzer **2012** and the controller **2014** determine positions, types, and the number of the game tokens **1L** bet by the losing player **2006L**, and calculate an increase in amount of the game token **1L**, which is the amount of the game token **1** expected to be in the collection chip tray **171**. Further, the image analyzer **2012** and the controller **2014** determine an actual total amount of the game tokens **1** in the collection chip tray **171** after the collection, and compare the expected total amount with the actual total amount to determine whether or not there is a difference.

[0110] The controller **2014** further performs redemption for the game token **1** to be redeemed, and after the redemption, compares a total amount of the chips expected to be in the chip tray based on the amount of the chips expected to be redeemed in the game, which is calculated from positions, types, and the number of chips each game participant bet at the game table and based on a game result, with the actual total amount of the chips in the chip tray to determine whether or not there is a difference.

[0111] Further, the game token **1W** is redeemed to the winning player **2006W** by using a game token **1** in a redemption chip tray **172**. Thus, it is possible to secure sufficient time for the image analyzer **2012** and the controller **2014** determine an actual total amount of the game tokens **1** in the collection chip tray **171** after the collection.

[0112] As described above, in the present embodiment, the controller **2014** calculates, for each game, income and outgo of chips from an amount of bet chips placed on the game table **2004** and a win/loss result of the game, and verifies an increase in balance of the chips in the chip tray **17** after the end of the game. If a difference is detected in this verification, the controller **2014** issues an alarm or adds a record of the difference to a record of a video image captured by the camera **2002**. A casino operator can track down a cause of the difference by reviewing the video image.

[0113] A table game management system according to the present embodiment calculates, from total amount of the game tokens **1** in the chip tray **17** before settlement for each game, an increase or decrease in amount of chips in the game calculated from positions, types, and the number of game tokens **1** bet by all players **2006** of the game and from a win/loss result of the game obtained by a win/loss result determination device. Then, the table game management system compares an expected total amount of the game tokens **1** in the chip tray **17** after settlement of the game end with an actual total amount of the game tokens **1** in the chip tray **17** at the game end, which is obtained via the image analyzer **2012**, to determine whether or not there is a difference.

[0114] Via the image analyzer **2012**, the controller **2014** determines positions, types, and the number of chips bet by each player, and determines an actual total amount of chips in the chip tray at an end of collection of all lost chips bet by each player. Then, the controller **2014** compares an expected total amount of the game tokens **1** in the chip tray **17**, which is obtained by adding an increase in chips in the chip tray **17** determined by positions, types, and the number of chips bet by a losing player, to a total amount of chips of the game in the chip tray before settlement of each game, with an actual total amount of the game tokens **1** in the chip tray **17** to determine whether or not there is a difference.

[0115] In a case where the controller **2014** compares an expected total amount of the game tokens **1** in the chip tray **17**, which is obtained by adding an increase in chips in the chip tray **17** determined by positions, types, and the number of the game tokens **1** bet by a losing player, to a total amount of the game tokens **1** of the game in the chip tray **17** before settlement of each game, with an actual total amount of the game tokens **1** in the chip tray **17** and determines there is no difference between the expected total amount and the actual total amount, and in addition, compares an expected total amount of the game tokens **1** in the chip tray **17** after settlement at the end of the game with an actual total amount of the game tokens **1** in the chip tray **17** at the end of the game, which is obtained via the image analyzer **2012**, and determines that there is a difference between the expected total amount and the actual total amount, the controller **2014** determines there has been a wrong payment and issues an error signal that notifies of the wrong payment.

[0116] The chip tray **17** includes the collection chip tray **171** that collects and temporarily keeps the game tokens **1** bet by a losing player. The image analyzer **2012** compares an expected total amount of the game tokens **1** in the collection chip tray **171**, which is obtained by adding an increase in amount of the game tokens **1** in the game calculated from positions, types and the number of the game tokens **1L** bet by the losing player, with an actual total amount of the game tokens **1** in the collection chip tray **171** to determine whether or not there is a difference between the expected total amount and the actual total amount.

[0117] In a case where the controller **2014** determines a difference in which a determined actual total amount of the game tokens **1** in the chip tray **17** of the dealer **2005** at the game table **2004** is not corresponding to an increase or decrease in amount of chips calculated from an amount of chips bet by all players and a win/loss result of the game, the game recorder **2011** can play back with an index or time attached to an acquired image or with a scene of collection or payment of the game token **1** specified, so that a record of the game in which the above different has occurred can be analyzed in the game recorder **2011**.

[0118] As described above, the controller **2014** obtains a total amount of chips in the chip tray **17** via the image analyzer **2012** after the settlement at the end of the game. In this case, determination after settlement is performed when: 1) redemption for the won game token **1** is completed; 2) the card C used in the game is collected and discarded in a discard area **41** or discard slot **42** of the table; 3) a predetermined button attached to a win/loss result determination device is pressed; or 4) a marker **43** that indicates win or loss is reset.

[0119] Further, the controller **2014** has an artificial-intelligence utilization structure or deep-learning structure, which allows the controller **2014** to determine positions (player, banker, or pair) and an amount (types and the number) of chips bet at each play position **7** of the game table **2004**, and to extract win/loss history of each player **2006**, which is obtained from a win/loss result of each game, and an amount of obtained chips (an amount of won chips) as an unusual situation specified by a casino, compared with statistical data of many games in the past (big data). Typically, the controller **2014** includes an artificial-intelligence utilization structure or deep-learning structure, which can extract, as an unusual situation compared with statistical data (big data, or the like) of games in the past, a case of a winning amount of a certain amount (one million dollars) or more, or a case of a series of several games in which amounts of bet chips are small in losing games and amounts of bet chips are large in winning games at a play position **7** of the game table **2004**.

[0120] Furthermore, unified with the image analyzer **2012**, the controller **2014** of this table game management system is configured to be able to identify an individual player **2006** at the play position **7** which has won a predetermined amount or more or at which an unusual situation is extracted. In identification of such a player **2006**, in the image analyzer **2012**, feature points of an image of a face is obtained by, for example, extraction of a feature point, and the image is identified by adding an identification number (ID, or the like) to the image. Then, the controller **2014** has an alarm function of notifying another game table of the presence of a specific player when the specific player **2006** leaves the seat and sits at the other game table. Specifically, a pit manager or a person in charge of each table (or a dealer) who manages each game table **2004** is notified to further prevent an unusual situation.

[0121] The controller **2014** can identify a collected game token **1** for each sitting number on the basis of information of an ID of RFID of the collected game token **1** determined by the chip tray determination device and information of an ID of RFID of the game token **1** read by the chip reader **25**.

[0122] In FIG. **11**, game tokens **1** are bet as illustrated in the detail tables in FIGS. **12** and **13**. In analysis by the image analyzer **2012**, places (player, banker, or the like) where the game tokens **1** are bet and the number (and amount) of the bet game tokens **1** can be determined. Meanwhile, in reading by the chip reader **25**, an ID of RFID attached to a game token **1** can be determined for each sitting number. Further, because a chip tray determination device **25** can determine IDs of game tokens **1** in a chip tray **9**, the chip tray determination device can determine a collected game token **1** by determining IDs added from before collection to after collection. In a case where a banker has won as a game result in FIGS. **11** to **13**, an analysis result by the image analyzer **2012** shows that, among IDs of the collected chips determined by the chip tray determination device, two pieces of chips bet on banker pair should be collected, and that two pieces of chips of ID **6** and ID **18** that the chip reader **25** has read as chips bet at the sitting number **5** have been collected.

[0123] The controller **2014** determines the total amount or the total number of the game tokens **1** expected to be collected for each sitting number, which is determined by using a measurement result of positions, an amount, and the number of the game tokens **1** based on a game result obtained from the game result determination device and an image analysis result by the image analyzer. Therefore, the controller **2014** can compare, to determine agreement, the total amount or the total number of the game tokens **1** collected for each sitting number, which is determined based on RFID, with the total amount or the total number of the game tokens **1** expected to be collected based on the image analysis result.

[0124] For example, in a case where a reading result by the chip reader **25** of the sitting number **1** is “A, B, C, D, E”, an analysis result by the image analyzer **2012** is “two pieces on tie, and two pieces on banker”, the game result is “winning of banker”, and a reading result by a chip tray determination device **18** is “A, B”, it is understood that the chips A and B bet on tie have been correctly collected.

[0125] The controller **2014** can determine whether or not information of an ID of RFID of the collected game token **1** determined by the chip tray determination device agrees with information of an ID of RFID of the bet game token **1** read by the chip reader **25**. By this determination, it is possible to determine whether or not the game token **1** collected in the chip tray is the game token **1** bet on the game table, and whether or not the game token **1** is wrongly collected from another place.

[0126] The controller **2014** can determine, on the basis of the RFID information read by the chip reader **25**, a chip type such as a cash chip, a rolling chip, or a promotion chip, and can store an amount of bet money for each type of the determined chips and an amount of money collected or redeemed.

[0127] A rolling chip is a special chip that can be purchased by a specific member and cannot be exchanged for cash, and an amount of the purchased rolling chips is managed for each member. A

rolling chip must be consumed for betting, and redemption for betting using a rolling chip is made by a cash chip. By storing an amount of the rolling chips bet or collected on the basis of the RFID information read by the chip reader **25**, an amount of rolling chips that a specific member is expected to have can be determined.

[0128] The chip reader **25** has a function of a confirmation area, and can read a game token **1** placed for redemption. As illustrated in FIG. **15**, in a case where redemption is made to banker at the sitting number **2**, the game token **1** for the redemption is placed and read in the chip reader **25**. In FIG. **15**, the game tokens **1** are placed next to the won game tokens **1** bet on banker at the sitting number **2**. The controller **2014** can compare an amount of redemption, which is determined by using a measurement result of a position, an amount, and the number of the game tokens **1** based on a game result obtained from the game result determination device and the image analysis result by the image analyzer, with an amount of the game tokens **1** placed for each sitting number, which is based on a reading result by the chip reader **25**, determine whether or not the amount of the redemption agrees with the amount of the game tokens **1** placed for each sitting number, and output a determined result. In FIG. **16**, at the sitting number **1**, the total number of the game tokens **1** expected to be paid is four, whereas a reading result by the chip reader **25** is three, which means the total number of the game tokens **1** and the reading result disagree. In a case where an amount of redemption disagrees with an amount of the game tokens **1** placed for each sitting number, a signal indicating the disagreement is output, and the dealer can replace a game token **1** for redemption on the basis of the output signal. In a case where the game token **1** is replaced, determination is performed again, and a determination result can be output. An area for placing the game token **1** for redemption may be anywhere in the chip reader **25**. Further, as illustrated in FIG. **15**, separately from a betting area, a payment confirmation area **26** may be arranged on a dealer side of the betting area. In FIG. **15**, a game token **1** for redemption is placed in a payment confirmation area **26** at a sitting number **6**, the chip reader **25** performs reading, the controller **2014** makes determination, and a redemption chip is placed next to the won game token **1** placed in banker.

[0129] For the confirmation area, the image analyzer **2012** may perform image analysis, and the controller **2014** may check a result of the image analysis with a reading result by the chip reader **25** to determine whether or not a total number of the result of the image analysis and a total number of the reading result by the chip reader **25** match.

[0130] On the basis of the determination made, the controller **2014** can output a signal. The signal may be displayed in a display unit of the chip tray **9** or a display unit of a card distributor **13**, or may be output to the pit manager or to the backyard of the casino.

[0131] The chip reader **25** may use an entire betting area as one reading area for each sitting number as illustrated in FIG. **17**, or may use, for each sitting number, a set of a player betting area and a banker betting area as one reading area, and a set of tie betting area, player pair betting area, and a banker pair betting area as one reading area as illustrated in FIG. **18**. Further, as illustrated in FIGS. **19** and **22**, each side betting area for a tie betting area, a player pair betting area, and a banker pair betting area may be treated as one reading area.

[0132] A game token **1** used in an embodiment will be described below. FIG. **20** is an external view of and FIG. **21** is a front sectional view of a game token **1** used in this system. The game token **1** has a multi-layer structure in which a plurality of plastic layers in different colors are laminated. The game token **1** includes at least an intermediate color layer **101** and white layers **102** or light color layers on both sides of the color layer **101**. The light color layers, which are not illustrated, are required at least to be layers in color lighter than color of the color layer **101**. With the multi-layer structure in which the intermediate color layer **101** and the white layers **102** or light color layers, which is not illustrated and required at least to be layers in color lighter than color of the color layer **101**, on both sides of the color layer **101** are laminated, as illustrated in FIG. **20**, to form a stripe pattern in a lamination direction on a side surface, and different colors (red, green, yellow, blue, or the like) are used for the color layer **101**, a type (10 points, 20 points, 100 points, 1,000

points, or the like) of the game token **1** can be identifiable.

[0133] As illustrated in FIG. **21**, the color layer **101** may be formed of a plurality of colored layers (FIG. **21** illustrates three layers). Because the plurality of the colored layers (three layers in FIG. **21**) are thermally press-bonded to one another, three-layer structure is not visible as illustrated in FIG. **21** that illustrates the three layers for description. Furthermore, a bore B is partially provided in a middle of the three layers of the color layer **101**, and RFID is embedded therein.

[0134] A structure of the game token **1** is not limited to a structure in which plastic is laminated, and may be similar to the color layer **101** in appearance.

[0135] A method for analyzing the game token **1** by the image analyzer **2012** will be described below in detail. The image analyzer **2012** analyzes an image recorded on a recorder **11** and recognizes at least two colors, which are a specific color **121** that is at least partially attached to the game token **1** and a reference color R that is present in the image and is different from the specific color **121**. Note that the specific color **121** is at least partially provided to a predetermined position of a game token **1** or in a predetermined form. For example, the specific color **121** may be provided as the color layer **101**, may be provided on a side surface of the game token **1** in a circumferential direction, or may be provided on a surface of the game token **1** as a predetermined mark. Further, the reference color R may be, for example, a color of a specific area of the game table **2004**, or may be a color attached to a place different from the specific color **121** in the game token **1**.

[0136] The image analyzer **2012** includes an artificial intelligence device **12a** that identifies a specific color by, for example, deep learning technology, and determines the number and types of the game tokens **1** placed on the game table **2004**. A recognition device **12** may further determine a position of the game token **1** on the game table **2004**.

[0137] The image analyzer **2012** acquires a plurality of images of the game token **1** and reference color R, which are irradiated with different illuminances, learns the acquired images and a correct specific color **121** and reference color R of the game token **1** in each image as teaching data by teaching from a human, and creates a learning model **13a** (recognition program). Note that a relative relation of the specific color **121** and the reference color R can be acquired, because in the images of the game token **1** and the reference color R irradiated with an illuminance of the same condition, the specific color **121** and the reference color R are irradiated with the same illuminance. For example, this relative relation may be used for recognition of the specific color **121**.

[0138] Further, without using artificial intelligence, a chip recognition device **12** may extract a center line C of the game token **1** with a method for measuring a feature of a shape, brightness, saturation, and hue in the images by using a result of an image analysis of the images captured by a camera **200212** and recorded as images.

[0139] Furthermore, the artificial intelligence device **12a** is configured to recognize at least two colors of the specific color **121** and the reference color R, which is different from the specific color **121** in a peripheral image of a predetermined range around the extracted center line C, by analyzing the peripheral image, which is for example a range of eight pixels orthogonal to the center line as a center. Note that, the extracted peripheral image of the predetermined range around the center line C may be analyzed as is, or after image processing such as color emphasis and noise reduction so that the specific color **121** can be easily recognized.

[0140] In summary, the artificial intelligence device **12a** of the recognition device **12** is an artificial intelligence device configured to extract the center line C from the image of the game token **1** and recognize at least two colors of the specific color **121** and the reference color R different from the specific color **121**, which are in the peripheral image, by analyzing the peripheral image of the predetermined range around the center line C as a center, and taught, as teaching data, the plurality of peripheral images of the game token **1** and the reference color R irradiated with different illuminances.

[0141] In another embodiment, the controller **2014** further includes a function to determine at least one among whether or not: 1) in each game, there is transference of a chip from time of start of

drawing cards or start of game operation by the dealer to time before display of a win/loss result of the game by a card distributor; 2) after an end of each game, there is transference of a chip by a person other than the dealer while the dealer is collecting the chips bet by a loser among the game participants; 3) after an end of each game, any chip has been added while the dealer is collecting a chip bet by a loser among the game participants; 4) after an end of each game, the dealer has paid to a position of a chip that a winner among the game participants has bet; and 5) after an end of each game, a winner among the game participants has taken a bet chip and a payed chip.

[0142] Further, the controller **2014** may be configured to determine at least one of 1) to 5) described above by detecting transference of hands of the dealer and game participant, transference of chips, or transference of the hands and the chips by using an analysis result from the image analyzer.

[0143] As a feature function, the controller **2014** has functions described in 1) to 5) below to determine whether or not there is any wrong against a rule of a baccarat game: [0144] 1) In each game, the controller **2014** monitors, by information that the image analyzer **2012** obtains by using the camera **2002**, whether or not there is transference of a game token **1** from time of a signal that starts drawing of cards obtained from the card distributor **2003**, or time of start of game operation by the dealer **2005** pressing a start button **4s** to time before display of a win/loss result of the game by the card distributor **2003**. [0145] 2) After an end of each game, the controller **2014** monitors, by information that the image analyzer **2012** obtains by using the camera **2002**, whether or not a loser **6** has wrongly taken a game token **1** while the dealer **2005** is collecting a game token **1** bet by a loser among game participants **2006**. [0146] 3) After an end of each game, the controller **2014** monitors, by information that the image analyzer **2012** obtains by using the camera **2002**, whether or not a person other than the dealer **2005** (winner or loser) has added a won game token **1W** or has replaced the game token **1** to a winning position not bet while the dealer **2005** is collecting a game token **1** bet by a loser among the game participants. [0147] 4) After an end of each game, the controller **2014** monitors, by information that the image analyzer **2012** obtains by using the camera **2002**, whether or not the dealer **2005** has correctly placed a won game token **1W** for payment to a position of the game token **1** that a winner among the game participants **2006** has bet. [0148] 5) After an end of each game (the dealer **2005** operates the card distributor **2003** to display a win/loss result in a display lamp **13**), the controller **2014** monitors, by information that the image analyzer **2012** obtains by using the camera **2002**, whether or not a winner **6W** among the game participants **2006** has taken a bet game token **1** and a payed game token **1W**.

[0149] Note that, in the above embodiment, the image analyzer **2012** and the controller **2014** are devices having an artificial-intelligence utilization structure or deep-learning structure. Specifically, the image analyzer **2012** and the controller **2014** may analyze an image or perform various controls described above by using scale-invariant feature transform (SIFT) algorithm, a convolutional neural network (CNN), deep learning, machine learning, or the like. These technologies perform image recognition of a captured image to recognize an object included in the image. In recent years, in particular, deep learning technology using a multilayered neural networks has been used to recognize an object with high accuracy. This deep learning technology generally recognizes an object with high accuracy by, over a plurality of stages, layering in an intermediate layer between an input layer and an output layer of a neural network. Regarding this deep learning technology, a convolutional neural network, in particular, has drawn attention for having higher performance than a conventional object recognizing method based on an amount of image features.

[0150] In a convolutional neural network, a recognition object image to which a label is attached is learned, and a main object included in the recognition object image is recognized. If there are a plurality of main objects in a learning image, learning is performed by designating a region with a rectangle and attaching a label to an image corresponding to the designated region. Furthermore, in a convolutional neural network, it is also possible to determine a main object and position of the object in the image.

[0151] To further describe a convolutional neural network, an object recognition process involves extraction of candidate regions based on a local feature by performing edge extraction processing, or the like, on a recognition object image, input of the candidate regions to the convolutional neural network, classification of the candidate regions after extracting feature vectors, and obtainment of a classified candidate region with a highest degree of certainty as a recognition result. A degree of certainty is quantity indicating how much similarity of an object in an image learned together with a certain image region and a label is relatively higher than similarity of another class.

[0152] As described above, various embodiments have been described. However, it is needless to say that a device of the present embodiments may be modified by a person skilled in the art as appropriate within the scope of the preset disclosure according to a game to which the present disclosure is applied.

[0153] In order to solve the above-described conventional problem, a game token management system according to an embodiment has a configuration in which: a single RFID tag having a data non-rewritable area and a data rewritable area is attached to a game token, the data non-rewritable area of the RFID tag stores, as fixed information, at least one of production information of the game token, product information of the game token, casino information, amount information of the game token, or a serial number of the game token, the data rewritable area stores, as variable information, location information of the game token, and the variable information stored in the data rewritable area can be extracted from the fixed information stored in the data non-rewritable area in an identical RFID tag, and the game token management system further includes: a writer that adds the location information of the game token to the data rewritable area, a reader that can collectively reads the fixed information stored in the data non-rewritable area of the plurality of game tokens and, on the basis of the read fixed information, can read the location information, which is a counterpart of the fixed information and stored in the data rewritable area, and a management controller that examines the location information obtained from the reader, in which the data rewritable area stores a plurality of pieces of location information including at least latest location information among all pieces of location information written by the writer, and the management controller includes a function to determine whether or not any abnormality is present in history of the location information obtained from the reader, and is configured to be able to, if there is abnormality, generate an error signal and identify the fixed information of the game token, which has been determined to be abnormal.

[0154] The data rewritable area is configured to store, together with the location information, information of time at which the writer writes to the game token, and the management controller is configured to determine, from the location information obtained from the reader, presence of abnormality by determining whether or not the game token has not been used for a predetermined period since last writing, or whether or not the location information, which is supported to be written, is not written.

[0155] In the management controller, the location information may include at least one of information indicating a cage, information indicating entry or exit, or information indicating a backyard, a situation where the game token has not been used for a predetermined period since last writing may be a situation where: 1) latest location information is information indicating a cage, and a predetermined period has passed before next information is written, 2) latest location information is information indicating entry, and a predetermined period has passed before next information is written, or 3) latest location information is information indicating exit, and a predetermined period has passed before next information is written, and a situation where location information, which is expected to be written, is not written may be a situation where: 4) at a time of entry to a gaming hall, information indicating exit is not written as latest location information in the data rewritable area of the game token, 5) at a time of use at the cage, information indicating a backyard is not written, or 6) at a time of exit from the gaming hall, information indicating a cage or a backyard is not written.

[0156] The location information may include information about an owner who owns the game token.

[0157] The management controller may have a function to determine, on the basis of the owner information obtained from the reader, presence of abnormality when: 1) a person different from a latest owner whose information is stored in the data rewritable area exchanges the game token for cash or another game token in the gaming hall, or exits the gaming hall, or 2) a person different from an owner at a time of exit whose information is stored in the data rewritable area brings the game token in the gaming hall.

[0158] The information about the owner of the game token may be identified by face authentication or ID card reading.

[0159] The location information may further include information indicating a place of a chip tray of a specific game table, information indicating a place of a betting area, or information indicating a payout area.

[0160] The management controller may have a function to determine, on the basis of information indicating location of the chip tray of the game table or information indicating a place of the betting area or of the payout area, which is obtained from the reader, presence of abnormality when: 1) a person different from a latest owner whose information is stored in the data rewritable area uses the game token at a game table, 2) a person different from the person who purchased a game token without a record of use at a game table attempts to exchange the game token at a cage, 3) a game token not including information indicating a cage or a backyard is used in the gaming hall, or 4) latest location information is information indicating a betting area or a payout area, and after a predetermined period has passed, the game token 1 is exchanged for cash at a cage or is used at a game table.

[0161] Information that the RFID tag has may be encrypted.

[0162] The data non-rewritable area may be configured to be functionally non-writable or locked to disable wiring of data.

[0163] The reader may be configured to collectively read the fixed information stored in the data non-rewritable area and location information stored in the data rewritable area, and the management controller may be configured to extract and determine, from the plurality of pieces of fixed information of the plurality of game tokens, which are obtained from the reader, the location information stored in the data rewritable area in an identical RFID tag.

[0164] In order to solve the above-described conventional problem, the game token according to an embodiment includes a single RFID tag having a data non-rewritable area and a data rewritable area, in which the data non-rewritable area stores, as fixed information, at least one of production information of the game token, product information of the game token, casino information, amount information of the game token, or a serial number of the game token, the data rewritable area is configured to store, as variable information, location information of the game token written by the writer, and configured to extract the variable information stored in the data rewritable area based on the fixed information stored in the data non-rewritable area in an identical RFID tag.

[0165] The game token may be configured to be able to collectively read the fixed information stored in the data non-rewritable area of the plurality of game tokens by a reader, and to read the location information stored in the data rewritable area on the basis of the read fixed information.

[0166] In order to solve the above-described conventional problem, the table game management system according to an embodiment includes: a game token to which an individually identifiable RFID is attached, a game table for playing a game by using the game token, a game recorder that records, via a camera, an image of a game token bet by each game participant of a game played at the game table, an image analyzer that performs image analysis on the recorded image of the game token, a chip reader that reads RFID attached to the game token bet by each game participant, and a controller that determines an image analysis result by the image analyzer and a reading result by the chip reader, in which the game table includes a plurality of betting areas including at least a

player betting area and a banker betting area for each sitting number, the chip reader reads the game token regarding an entire betting area as one reading area for each sitting number, or reads the game token for each of a plurality of reading areas by dividing the entire betting area into a plurality of reading areas which are less than the number of betting areas, and the controller is configured to be able to determine types and the number of the bet game tokens separately for each betting areas including at least the player betting area and the banker betting area, on the basis of an image analysis result by the image analyzer, determine whether or not a total amount or the total number of the game tokens for each of the sitting numbers or for each of the reading areas based on an image analysis result by the image analyzer agrees with a total amount or the total number of the game tokens for each of the sitting numbers or for each of the reading areas based on a reading result by the chip reader, and output a determination result.

[0167] The controller may be configured to be able to output a sitting number with a determination result of disagreed amount or number.

[0168] The controller may be configured to be able to generate a signal that indicates a possibility that RFID attached to the game token is broken or RFID is not attached to the game token in a case where a total amount or the total number of the game tokens for each of the sitting numbers based on a reading result of the chip reader is smaller than a total amount or the total number of the game tokens for each of the sitting numbers based on an image analysis result by the image analyzer.

[0169] The controller may further be configured to be able to generate a signal that indicates a possibility of a reading error by the image analyzer reading the game token in a case where the total amount or the total number of the game tokens for each of the sitting numbers based on a reading result of the chip reader is larger than the total amount or a total number of the game tokens for each of the sitting numbers based on an image analysis result by the image analyzer.

[0170] The table game management system may include a database storing an ID of RFID attached to the game token, and the controller may be configured to be able to check an ID of RFID based on a reading result by the chip reader with an ID stored in the database, determine authenticity of the game token, and output a determination result.

[0171] The table game management system includes: a chip tray that keeps a game token used by a dealer at a game table, a chip tray determination device that can determine types and the number of the game tokens contained in the chip tray, and a game result determination device that determines a game result of each game at the game table, in which the controller may include: a calculation function that calculates income and outgo at a game table for a casino, at every collection and redemption for each game by using a measurement result of positions, an amount, and the number of game tokens based on a game result obtained from the game result determination device and an image analysis result by the image analyzer, a collection determination function that, after collection of lost chips bet by each game participant, compares an amount of chips at the game expected to be collected, which is calculated from positions, types, and the number of chips bet by each game participant at the game table, and the game result with an amount of actually collected chips in the chip tray, and determines whether or not there is a difference between the amount of the chips expected to be collected and the amount of the actually collected chips to determine whether or not there has been any wrong or an error with the collection of the game token, and a redemption determination function that performs redemption for a chip expected to be redeemed, compares, after the redemption, a total amount of chips expected to be in the chip tray based on an amount of the chips to be redeemed in the game, which is calculated from positions, types, and the number of chips bet by each game participant at the game table, and the game result, with an actual total amount of chips in the chip tray, and if having determined that there is a difference between the expected total amount and the actual total amount, determines that there has been a wrong or an error with the redemption.

[0172] The controller may be configured to identify the collected game token for each sitting number on the basis of information of an ID of RFID of the collected game token determined by

the chip tray determination device and information of an ID of RFID of the game token read by the chip reader, compare, IDs, an amount, or the number of the identified game tokens with IDs, an amount, or the number of the game tokens expected to be collected for each sitting number, which has been determined by using a measurement result of positions, an amount, or the number of the game tokens based on a game result obtained from the game result determination device and an image analysis result by the image analyzer, and determine whether or not the IDs, an amount, or the number of the identified game tokens agrees with the IDs, an amount, or the number of the game tokens expected to be collected for each sitting number.

[0173] The controller may be configured to be able to determine, at each game, a type of the game token classified by application or purpose of use on the basis of information of ID of RFID of the game token read by the chip reader, and store, for each sitting number, an amount of the bet game tokens or an amount of the collected or redeemed game tokens by the type.

[0174] The controller may be configured to be able to store a cumulative amount of the game tokens bet by type or a cumulative amount of the collected or redeemed game tokens for each sitting number in a plurality of games.

[0175] A type of the game token may be at least a cash chip, a rolling chip, or a promotion chip.

[0176] The controller may be configured to be able to identify an owner on the basis of ID information of the rolling chip and store a cumulative amount of bet, collected, or redeemed rolling chip for each owner.

[0177] The chip reader may be configured to read the game token for redemption placed on the chip reader for each sitting number, and the controller may be configured to be able to compare an amount of redemption, which is determined by using a measurement result of a position, an amount, and the number of the game tokens based on a game result obtained from the game result determination device and an image analysis result by the image analyzer, with an amount of the game tokens placed for each sitting number, which is based on the reading result by the chip reader, determine whether or not the amount of the redemption agrees with the amount of the game tokens placed for each sitting number, and output a determined result.

[0178] Further, the controller may be configured to be able to display disagreement in a case where a determination result of the redemption disagrees, perform determination again in a case where there has been any change in amount of the game tokens placed on and read by the chip reader for each sitting number, and output agreement or disagreement again.

[0179] The controller may be configured to be able to compare an amount of the game tokens placed for redemption for each sitting number with increase or decrease in amount of the game tokens in the chip tray from before redemption to after the redemption and determine whether or not there is a difference.

[0180] The chip reader may include a payment confirmation area, which is arranged on a dealer side of the betting area, for placing the game token for redemption.

[0181] The table game management system may further include a game participant identification unit using a face authentication system or a player ID card, and the controller may be configured to be able to associate the sitting number with information of the game participant identified by the game participant identification unit.

[0182] A display unit that displays based on output of the controller may be included.

[0183] The game table may include a side betting area, in which the side betting area may be adjacently provided with a betting area for each sitting number, the chip reader may be able to collectively read at least one side betting area, and the controller may be configured to be able to determine types and the number of the game tokens bet at least one side betting area on the basis of an image analysis result by the image analyzer, determine whether or not a total amount or the total number of the game tokens in at least one side betting area based on an image analysis result by the image analyzer agrees with a total amount or the total number of the game tokens in the side betting area based on a reading result by the chip reader, and output a signal of a determination

result.

[0184] Another embodiment is a table game management system including: a game token to which an individually identifiable RFID is attached, a game table for playing a game by using the game token, a game recorder that records, via a camera, an image of a game token bet by each game participant of a game played at the game table, an image analyzer that performs image analysis on the recorded image of the game token, a chip reader that reads RFID attached to the game token bet by each game participant, and a controller that determines an image analysis result by the image analyzer and a reading result by the chip reader, in which the game table includes a betting area including a player betting area, a banker betting area, and a side betting area, the side betting area is adjacently provided with a betting area for each sitting number, the chip reader is able to collectively read at least one side betting area, and the controller is configured to be able to determine types and the number of the bet game tokens separately for at least one side betting area on the basis of an image analysis result by the image analyzer, determine whether or not a total amount or the total number of the game tokens for at least one side betting area based on an image analysis result by the image analyzer agrees with a total amount or the total number of the game tokens of the side betting based on a reading result by the chip reader, and output a signal of a determination result.

[0185] The controller may be configured to be able to output the side betting with a determination result of disagreeing amount or number.

[0186] The controller may be configured to be able to generate a signal that indicates a possibility that RFID attached to the game token is broken or RFID is not attached to the game token in a case where a total amount or the total number of the game tokens for each side betting based on a reading result of the chip reader is smaller than a total amount or the total number of the game tokens for each side betting based on an image analysis result by the image analyzer.

[0187] The controller may be configured to be able to generate a signal that indicates a possibility of a reading error by the image analyzer reading the game token in a case where a total amount or the total number of the game tokens for each side betting based on a reading result of the chip reader is larger than a total amount or the total number of the game tokens for each side betting based on an image analysis result by the image analyzer.

[0188] The controller may include a database storing ID of RFID attached to the game token and may be configured to be able to check an ID of RFID based on a reading result by the chip reader with an ID stored in the database, and determine authenticity of the game token.

[0189] The controller may be configured to be able to identify the collected game token for each side betting on the basis of information of an ID of RFID of the collected game token determined by the chip tray determination device and information of an ID of RFID of the game token read by the chip reader, compare IDs, an amount, or the number of the game tokens expected to be collected for each side betting, which has been determined by using a measurement result of positions and an amount of the game tokens based on a game result obtained from the game result determination device and an image analysis result by the image analyzer, with IDs, an amount, or the number of the game tokens collected for each side betting, and determine whether or not a collecting amount agrees for each side betting.

[0190] If there is, in an area of one chip reader **25**, one chip or one chip stack to which payout chip is to be paid, the bet chip and the payout chip can be paired and associated with owner information without detecting the movement of the chip out of the area of the chip reader **25**. That is, in the case where there is one bet chip or one stack consisting of a plurality of bet chips **1001** in an area of one chip reading device **25**, even if it is not detected that the chip has moved outside the area of the chip reading device **25**, the owner information can be associated by pairing the bet chip and the payout chip based on the information in which the payout chip has disappeared from the chip tray or the information in which the payout chip has been read by the chip reading device **25** on the table. Of course, the bet chip may be recognized on the basis of the information that the chip has

moved out of the area of the chip reading device 25, and the payout chip and the bet chip may be paired based on the owner information of the bet chip, by using some of the information among the information on the chip taken out from the chip tray, the information on the chip placed in the betting area, and the information on the chip moved out of the area of the chip reading device 25. Alternatively, the ID of the chip may be read by a RFID, and the type and the number of chips, also the identification of stacks may be recognized by cameras.

[0191] The case in which there is one chip or one chip stack to which payout chip is to be paid in an area of one chip reader 25 includes a case in which there was one chip or one chip stack to which payout chip is to be paid in the area at a timing of betting and a case in which there was a plurality of chip stacks at a timing of betting but only one stack remained in the course of proceeding with payout. Only the remaining one bet chip stack may be paired with the payout chip to be associated with the owner information. In this case, by treating the remaining one bet chip stack as the chip bet by the member, tracing of chips of members can be ensured. It is preferable that, only one member player may bet at each chip reader (e.g., each player position) and a plurality of non-member players may bet at each chip reader. In the case where there is one member player for each player position, since the chip and the owner are associated with each other, the information of the member player assigned for each player position can be determined from the information of the bet chip. Also, if there is a rule that a member player shall bet a chip prior to a non-member player, the first bet chip may be associated with the member player information assigned to the player position (e.g., sitting at the player position 1), or such an association may be confirmed. A member card may be used to assign member players to player positions.

[0192] The chip reading device 25 may be provided for each area such as a player, a bunker, a tie, a pair, or the like, or may be a large reading device 25 may be provided for each player position, for each divided areas which is obtained by dividing the whole table into a plurality of areas such as an area including the player positions 1 to 3 and an area including the player positions 4 to 6, or for whole table. In any case, the bet chip and the payout chip can be paired and the owner information and the like can be updated based on information that the chip has moved out of the area of the chip reading device 25 (that the chip cannot be read by the chip reading device 25), the ID of the payout chip that has disappeared from the chip tray, the ID of the payout chip placed in the bet area, and the like.

Claims

1. A table game management system for a game table for playing game with a gaming chip with an RFID storing a chip ID that can be individually identified, the table game management system comprising: a bet chip determination unit configured to identify the chip ID of a bet chip by reading the RFID of the bet chip placed in a betting area; a payout chip determination unit configured to identify the chip ID of a payout chip to be paid from chip tray to a bet chip that won the game; a memory configured to store the chip ID of the gaming chip and an owner information of the gaming chip in association with each other; and an updating unit configured to rewrite an owner information associated with the chip ID of the payout chip to be paid to the bet chip that won game in the memory, to an owner information of the bet chip that won the game, wherein, when a plurality of players place bet chips in the same bet area: the bet chip determination unit is configured to identify a chip ID of last remaining bet chip by reading the RFID of the last remaining bet chip, after the bet chip of winning player among the plurality of players and payout chips paid to them have been transferred outside the bet area first, and the updating unit is configured to rewrite the owner information associated with the chip ID of the payout chip identified by the payout chip determination means in the memory to the owner information associated with a chip ID of the last remaining bet chip, and wherein the payout chip determination unit is configured to: (1) identify the chip ID stored in the RFID, which is no longer being read in

the chip tray by removing the payout chip from the chip tray, as a chip ID of the payment chip; (2) identify a chip ID of the payout chip by reading, in a confirmation area of the gaming table, the RFID of the payout chip that has been removed from the chip tray and placed in the confirmation area; (3) identify a chip ID of the payout chip by reading the RFID of the payment chip by a reader for the RFID of the gaming chip bet in the betting area; or (4) identify the chip ID stored in the RFID, which is no longer read in the betting area due to the player taking the payment chip from the betting area, as a chip ID of the payment chip.

2. The table game management system according to claim 1, further comprising a player identification unit configured to identify player information about who is playing a game at the game table using face authentication technology or an ID card.

3. The table game management system according to claim 2, wherein the updating unit is further configured to determine that fraud exists when a combination of the player information identified by the player identification unit and a chip ID of gaming chips that the player has bet in the betting area does not match a combination of a chip ID of the gaming chip stored in the memory and owner information associated with a chip ID of the gaming chip.
